Implementing Privacy-Preserving Techniques for Secure Healthcare Data Analytics

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Objectives

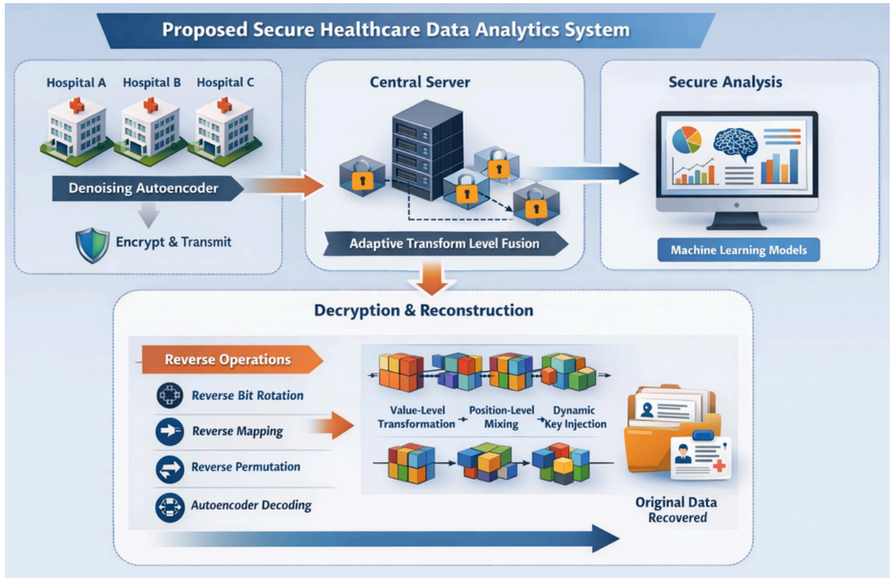
The main objective of this project is to develop a privacy-preserving healthcare data analytics framework that allows multiple hospitals to securely share and analyze medical data while protecting sensitive patient information. Healthcare datasets contain confidential details such as patient records, diagnosis information, and treatment history, which must be protected during storage, transmission, and analysis. To achieve this, the system implements a two-level security mechanism using Denoising Autoencoder and Adaptive Transform Level Fusion encryption to protect both the structural and statistical properties of the data. Through this approach, hospitals can share encrypted datasets with a central server where secure data fusion and machine learning analytics can be performed without revealing the original patient information, ensuring both effective healthcare analytics and strong patient data privacy.

Existing System

In traditional healthcare data analytics systems, datasets from different hospitals are collected and stored in a centralized server for analysis. These systems often rely on basic encryption techniques or simple data protection methods. However, during data sharing and analytics processes, sensitive patient information may still be exposed. Existing systems usually provide single-layer security, which is not sufficient to prevent statistical attacks or pattern-based data leakage. Additionally, these systems do not effectively hide the structural relationships within the datasets, making them vulnerable to privacy breaches. As a result, there is a need for a more advanced privacy-preserving framework that can secure healthcare data while still enabling effective analytics.the centralized storage of large-scale healthcare data increases the risk of unauthorized access and data misuse. If the central server is compromised, the entire dataset containing confidential patient information may be exposed.

Proposed System

The proposed system introduces a two-level privacy-preserving architecture for secure healthcare data analytics across multiple hospitals. In the first level, each hospital processes its dataset using a Denoising Autoencoder, which removes noise and encodes the dataset into a latent representation, providing structural protection. The encoded data is then encrypted and transmitted securely to a central server. In the second level, encrypted datasets from different hospitals are combined using Adaptive Transform Level Fusion, which applies value-level transformation, position-level mixing, and dynamic key injection to enhance security. During data retrieval, the system performs decryption using reverse operations such as reverse bit rotation, reverse nonlinear mapping, reverse permutation, and autoencoder decoding. This dual-layer approach ensures high data privacy while enabling secure machine learning analytics.



System Requirements

Hardware Requirements

The system requires a computer with a minimum Intel i5 or equivalent processor, 8 GB RAM, and sufficient storage capacity such as 256 GB SSD to handle dataset processing and machine learning computations.

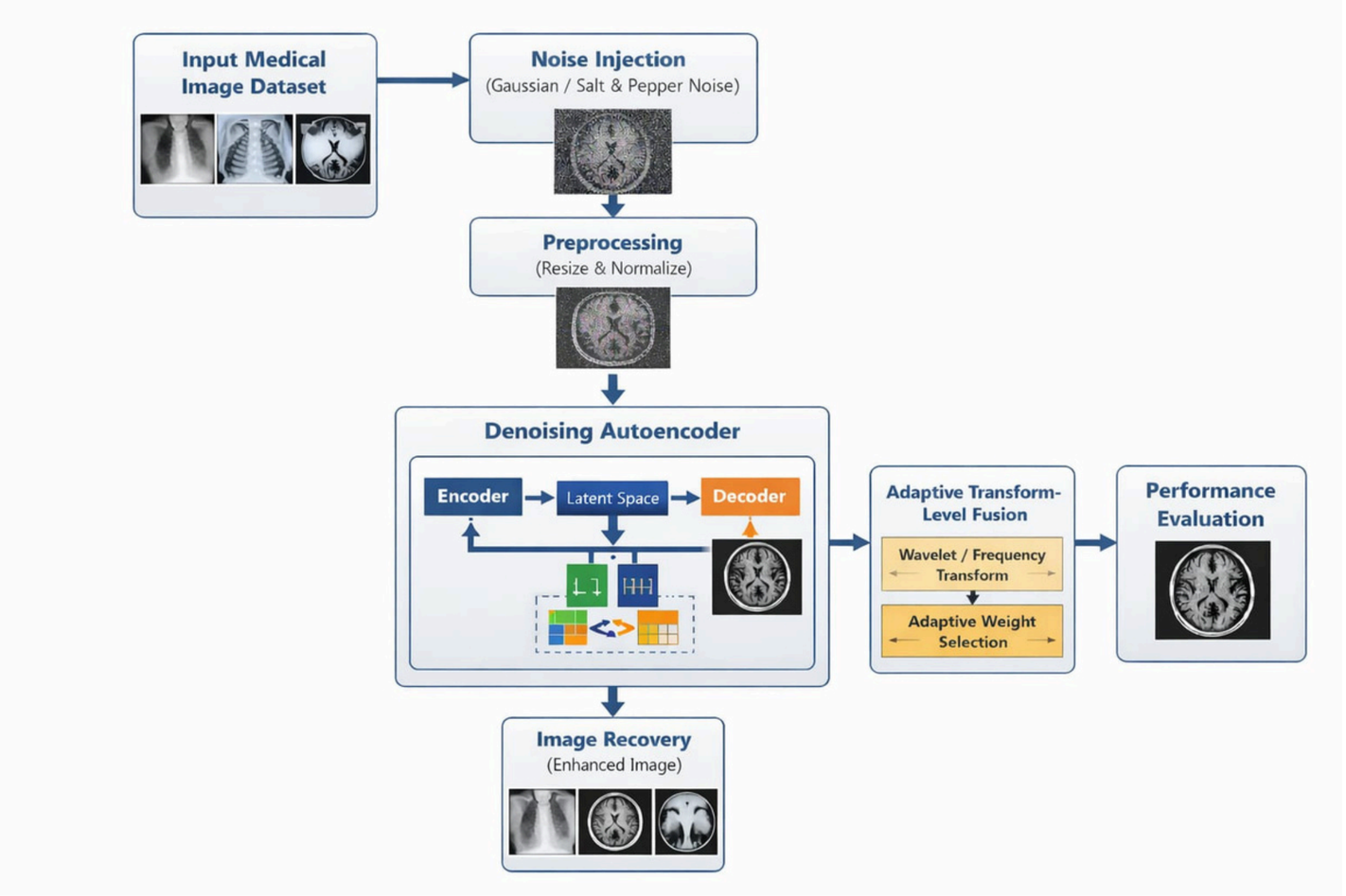
Software Requirements

The software environment includes Python programming language for implementing machine learning models and encryption algorithms. Development tools such as Jupyter Notebook or Visual Studio Code are used for coding and experimentation. Libraries such as TensorFlow, NumPy, Pandas, Scikit-learn, and Matplotlib are used for data processing, model development, and visualization.

Literature Survey

S.No	TITLE	AUTHOR(S)	INFERENCE	PUBLISHED IN
1.	Privacy Preservation In Artificial Intelligence - Enabled Healthcare Analytics	Shancang Li1* , Muddesar Iqbal2 , Ali Kashif Bashir3,5,6 and Xinheng Wang4	privacy-preserving methods improve secure data storage and controlled access in healthcare systems.	RESEARCH (Research Manuscript) Open Access
2.	Privacy - Preserving AI For Secure Data Sharing In Healthcare	Rajashekhar Reddy Kethireddy	encryption-based privacy techniques can securely enable healthcare data analysis without compromising patient confidentiality	Journal of Recent Trends in Computer Science and Engineering (JRTCSE)
3.	AI- Driven Secure Data Sharing: A Trustworthy And Privacy - Preserving Approach	Al Amin∗ , Kamrul Hasan∗ , Sharif Ullah† , Liang Hong∗ ∗Tennessee State University, TN, USA	encrypted data can be effectively used in advanced machine learning models while maintaining high privacy and accuracy.	University of Central Arkansas, Conway, AR, USA
4.	Privacy-Preserving Machine Learning For Healthcare: Open Challenges And Future Perspectives	Alejandro Guerra-Manzanares∗ , L. Julian Lechuga Lopez,∗ Michail Maniatakos andFarah E.Shamout	combining security mechanisms with machine learning enhances reliable and safe healthcare predictions.	ICLR 2023 Workshop on Trustworthy Machine Learning for

Architecture Diagram:



Modules Completed

1. Data Collection Module (COMPLETED)

Healthcare datasets are collected from multiple hospitals for secure analytics.

2. Denoising Autoencoder Module (COMPLETED)

Each hospital processes its dataset using a Denoising Autoencoder to remove noise and generate a secure encoded representation.

3. Encryption Module (COMPLETED)

The encoded data is encrypted to protect structural information before transmission.

4. Secure Data Transmission Module

Encrypted datasets are securely transmitted from individual hospitals to the central server.

5. Adaptive Transform Level Fusion Module

Encrypted datasets from different hospitals are fused using adaptive transform level fusion techniques.

6. Decryption and Reconstruction Module

The encrypted dataset is decrypted using reverse operations, and the Autoencoder decoder reconstructs the data for further analytics.

Work Plan & Timeline

Phase	Activities	Duration	Phase	Activities
Phase 1	Problem Definition & Literature Survey	2 weeks	Phase 1	Identifying privacy issues in healthcare data sharing and studying existing research works.
Phase 2	Datasett collection and model development	4 weeks	Phase 2	Collecting healthcare datasets and implementing the Denoising Autoencoder model for data encoding.
Phase 3	Encyption And Data Fusion Implementation	4 weeks	Phase 3	Applying encryption techniques and performing Adaptive Transform Level Fusion to combine encrypted datasets securely.
Phase 4	Decryption, Evaluation ,and documentation	1 weeks	Phase 4	Reconstructing the dataset using the decoder, evaluating security performance, and preparing the final report and presentation.

Output

Original Healthcare Dataset

	Attribute 1	Attribute 2	Attribute 3	Attribute 5
1	0.67	0.25	0.71	0.84
2	0.52	0.15	0.62	0.92
3	0.76	0.81	0.39	0.01
4	0.34	0.13	0.82	0.93
5	0.12	0.47	0.25	0.95
5	0.12	0.47	0.76	0.55

Encrypted Cipher Dataset

	Attribute 1	Attribute 2	Attribute 3	Attribute 4
1	-YJbP -	oVhd	cDH3	cv7y
2	Dqo8	kB2H	n0ag	p6FZ
3	tNS1	MLfk	2X1R	LxYJ
4	1ZSp	Cfoa	sZEu	KZ4j
5	uETf	vQDI	NCjy	iQ46
5	uETf	vQDI	NCJ6	xvdP

Reconstructed Dataset After Decryption

	Attribute 1	Attribute 2	Attribute 5	Attribute 5
1	0.68	0.25	0.74	0.83
2	0.52	0.14	0.62	0.93
3	0.74	0.78	0.30	0.01
4	0.34	0.12	0.62	0.91
5	0.13	0.46	0.25	0.75
5	0.13	0.46	0.25	0.56

Conclusion

- This project works on secure dataset processing to improve the quality and reliability of data.
- The Denoising Autoencoder helps remove noise from the dataset and produces a cleaner output.
- The Adaptive Transform-Level Fusion method enhances important information and improves the final result.
- The proposed system helps maintain data security and integrity while processing the dataset.
- Overall, the system produces a better quality and more secure output dataset, which can be useful for further analysis and applications.

Future Work

- The proposed system can be extended to work with larger and more diverse datasets to improve performance and reliability.
- Advanced deep learning models can be integrated to further enhance noise removal and feature extraction.
- The system can be improved to support real-time data processing for faster and more efficient results.
- Additional security techniques and encryption methods can be incorporated to provide stronger protection for sensitive datasets.
- Future research can focus on optimizing computational efficiency to reduce processing time and resource usage.

THANK YOU